12th HACKERS KA JUGAAD

Aldehydes, Ketones & Carboxylic Acid



carbonyl group.

↳ hybridisation = π x
σ + LP
4 sp^3
3 sp^2
2 sp

↳ RCOOH or $R-\overset{\overset{O}{\|}}{C}-OH$

aldehyde → R-CHO or $R-\overset{\overset{O}{\|}}{C}-H$

$H-\overset{\overset{O}{\|}}{C}-H$ or HCHO

Ketones



or RCOR.

IUPAC naming

Side chain + extra grp ↖ 2° prefix + 1° prefix + word Root + 1° Suffix + 2° Suffix

used for cyclic ↳ cyclo

C-C - ane
C=C - ene
C≡C - yne

aldehyde - al
ketone = one

+ Senior most func. grp.

↳ total no. of carbon atoms

formyl (or oxo) - aldehyde not in main chain

CHO grp is oxo or keto - ketone not part of main chain

C_1 = meth
C_2 = eth
C_3 = prop
C_4 = but
C_5 = pent
C_6 = hex
C_7 = hept
C_8 = oct
C_9 = non
C_{10} = dec
C_{11} = undec
C_{12} = dodec

MOP of both Aldehydes, Ketones

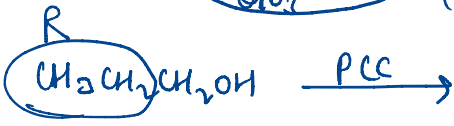
1) By oxidation of Alcohols

1° alcohol $RCH_2OH \xrightarrow{KMnO_4}$ aldehyde $\longrightarrow$ $RCOOH$ carboxylic acid.

$RCHO$

PCC (pyridinium chlorochromate) → good yield.

Mild CrO_2 (poor yield)



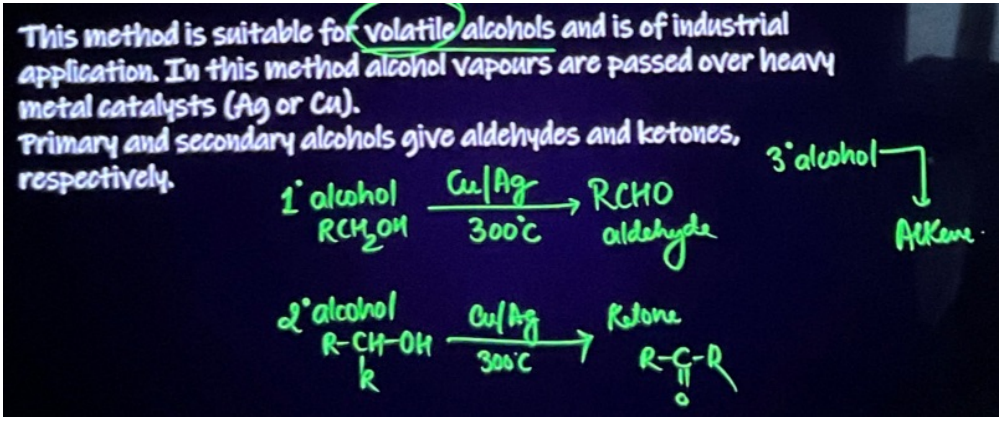
$\xrightarrow{PCC}$ $RCHO$

CH_3CH_2CHO

2° alcohol $\xrightarrow{\text{strong acid}}$ Ketone



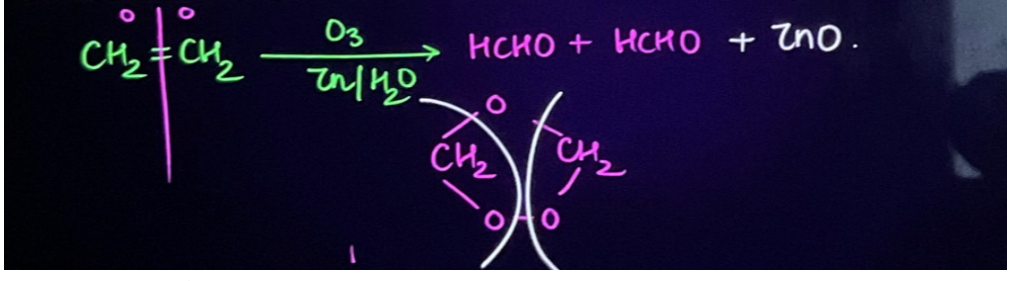
2) By dehydration of Alcohol -



3) From Hydrocarbons -

(i) By ozonolysis of alkenes : Reductive Ozonolysis

Ozonolysis of alkenes followed by reaction with zinc dust and water gives aldehydes, ketones or a mixture of both depending on the substitution pattern of the alkene.



$CH_3-CH=CH_2 \xrightarrow[Zn/H_2O]{O_3} CH_3CHO + HCHO + ZnO$

$CH_3-\underset{CH_3}{C}=CHCH_3 \xrightarrow[Zn/H_2O]{O_3} CH_3COCH_3 + CH_3CHO$

or $CH_3-\underset{\underset{O}{\|}}{C}-CH_3$

4.) Thermal decomposition of Monocarboxylic acid

$2HCOOH \xrightarrow{\Delta} H_2O + CO_2 + HCHO$

HCOOH
HCOOH

$HCOOH + RCOOH \xrightarrow{\Delta} H_2O + CO_2 + RCHO$

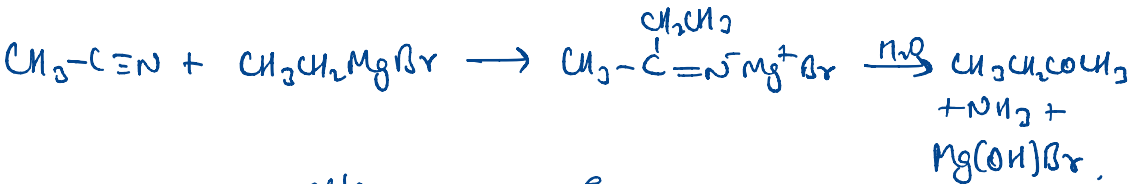
$CH_3COOH + CH_3COOH \xrightarrow{\Delta} H_2O + CO_2 + CH_3COCH_3$

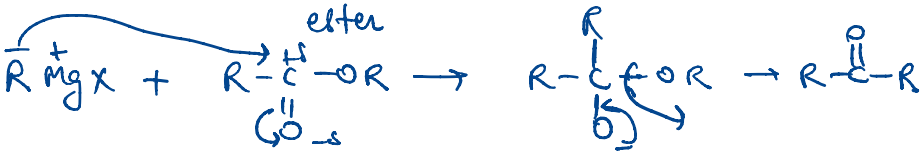
Grignard reagent

$R\overset{+}{M}gX + R-\overset{+\delta}{C}\equiv\overset{-\delta}{N} \longrightarrow R-\underset{R}{C}=\overset{\ominus}{N}\overset{\oplus}{M}gX \xrightarrow[H^+/OH]{\textcircled{2}\ H_2O} NH_3 + Mg(OH)X$

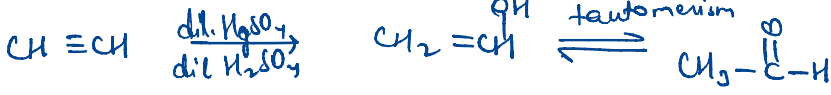
$R-\overset{O}{\overset{\|}{C}}-R$

or RCN

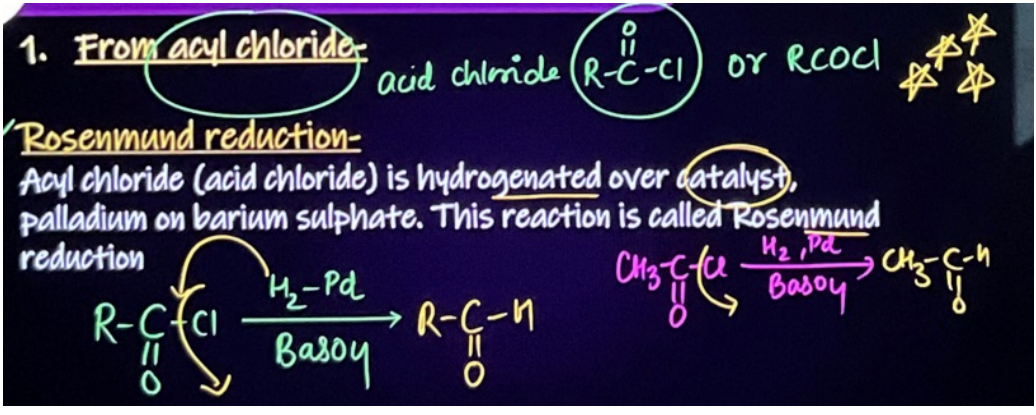


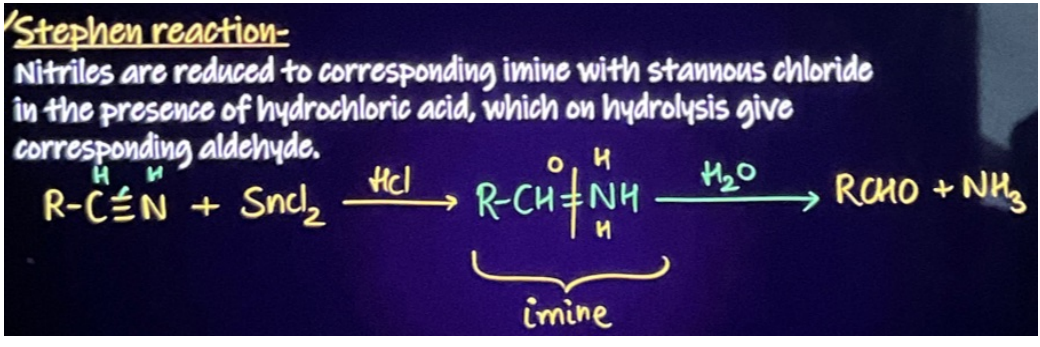


Hydration of alkynes



MOP of only Aldehyde





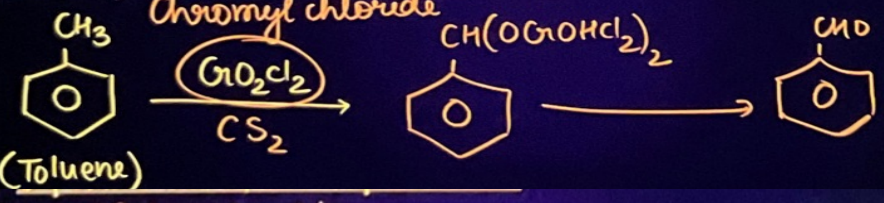
3. From hydrocarbons-

ETARD REACTION

A) By oxidation of methylbenzene-

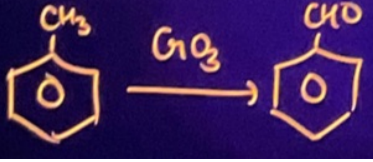
a) Use of chromyl chloride

Chromyl chloride oxidises methyl group to a chromium complex, which on hydrolysis gives corresponding benzaldehyde.

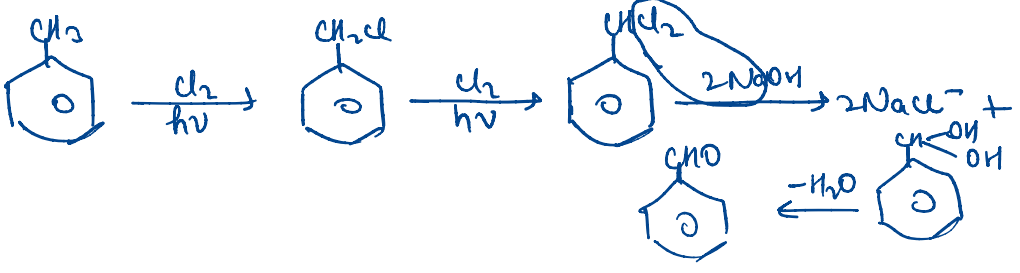


b) Use Of chromic oxide-

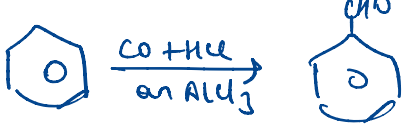
Toluene or substituted toluene is converted to benzylidene diacetate on treating with chromic oxide in acetic anhydride. The benzylidene diacetate can be hydrolysed to corresponding benzaldehyde with aqueous acid.



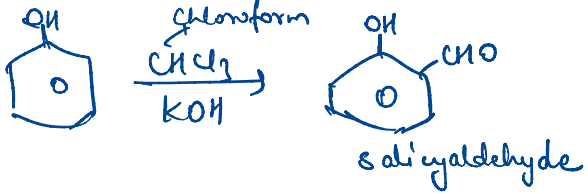
Side chain chlorination



Grattermann Koch synthesis



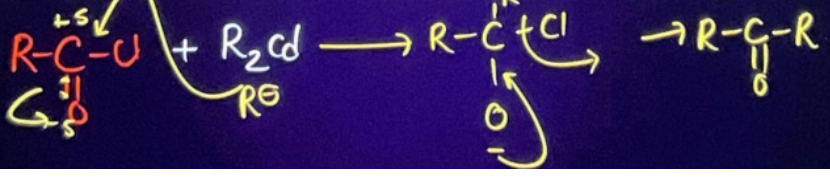
Reimer- Teimann Rx



MOP of only Ketones

1. From acyl chloride-

Treatment of acyl chlorides with dialkylcadmium, prepared by the reaction of cadmium chloride with Grignard reagent, gives ketones



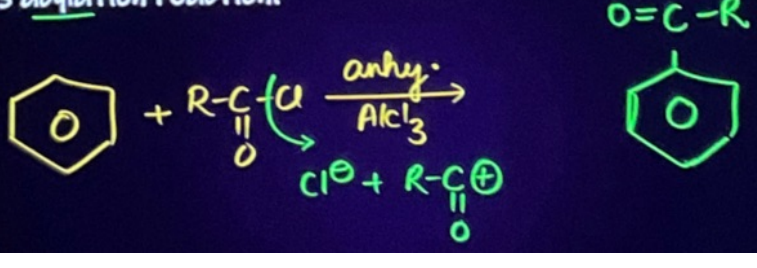
2. From nitriles-

Treating a nitrile with Grignard reagent followed by hydrolysis yields a ketone.

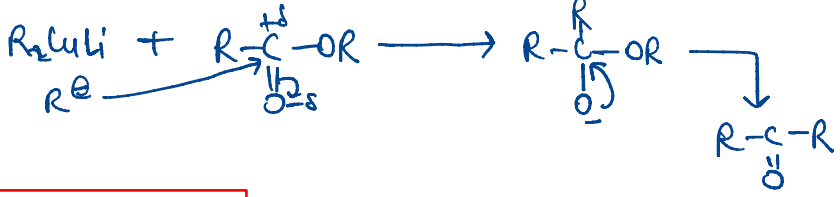
Already done.

3. From benzene or substituted benzene-

When benzene or substituted benzene is treated with acid chloride in the presence of anhydrous aluminium chloride, it affords the corresponding ketone. This reaction is known as Friedel-Crafts acylation reaction.



Gilman's Reagent

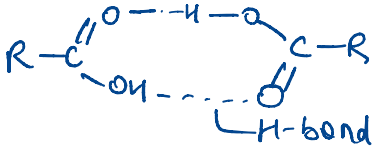


Physical properties

1.) Bpt $\propto$ mol. mass
$\propto$ no. of carbons
$\propto \frac{1}{\text{branching}}$

Carboxylic acid > Alcohol > Aldehyde ketones > ether > hydrocarbon.

- H bond
- dimer

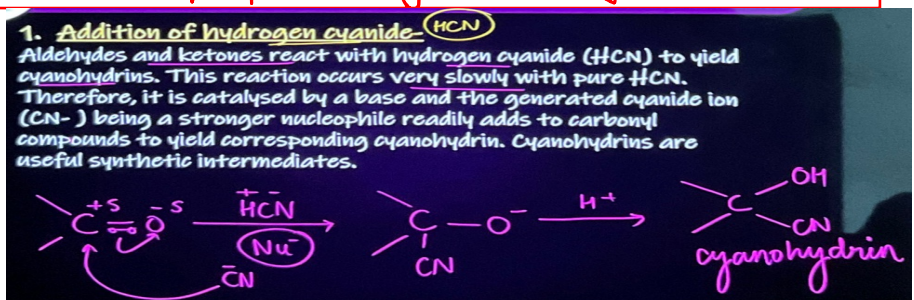


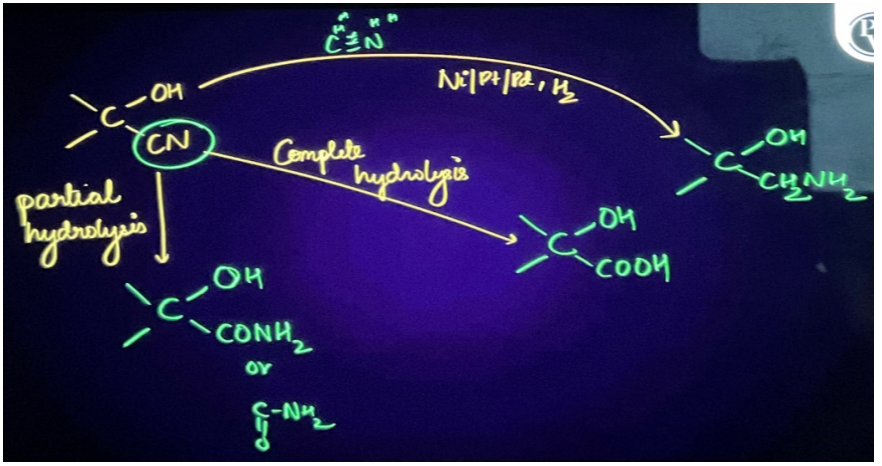
2.) Solubility –

Lower Molecules soluble in H_2O
↳ C-1, 2

C↑↑↑, hydrophobic part ↑ solubility ↓↓
H_2O hate.

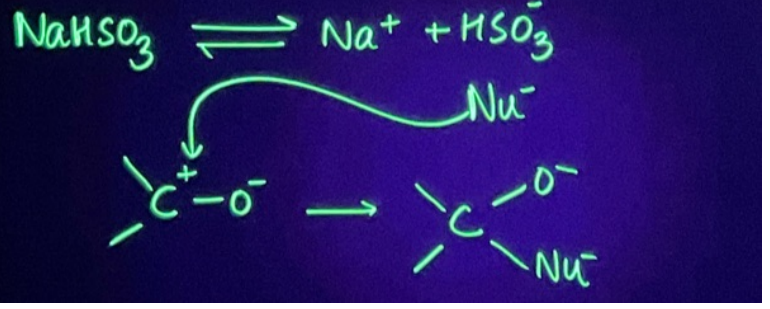
Chemical properties of Aldehydes, Ketones



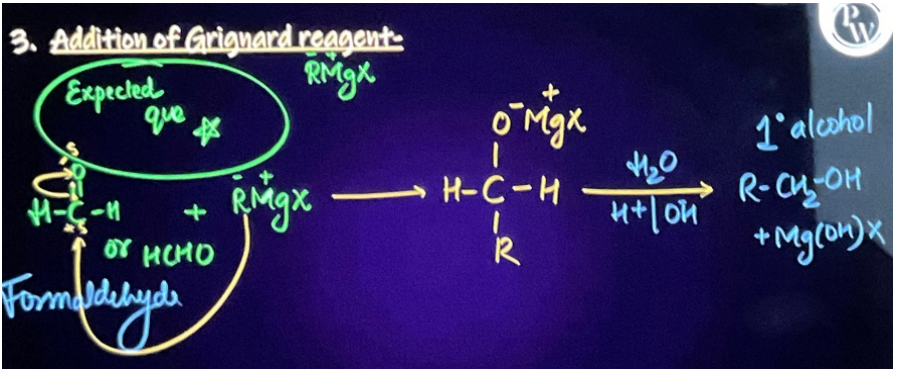


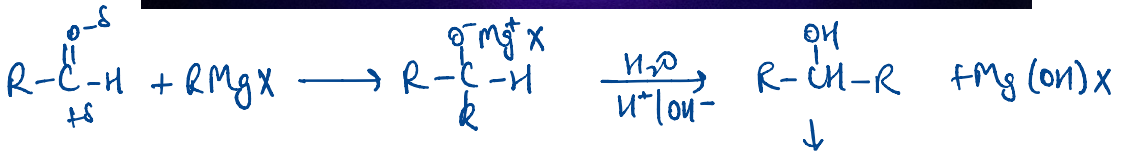
2. Addition of sodium hydrogensulphite-

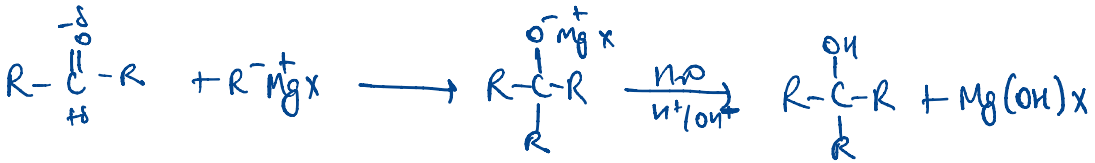
Sodium hydrogensulphite adds to aldehydes and ketones to form the addition products.



3. Addition of Grignard reagent-



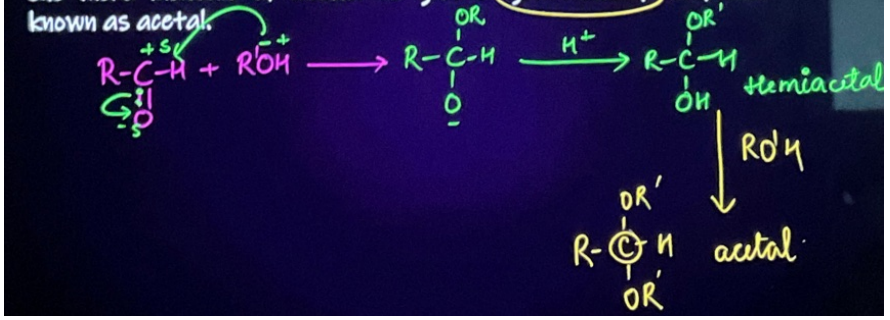


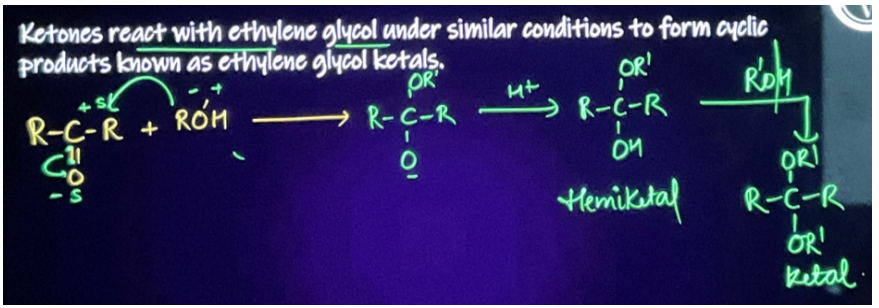


3° Alcohol.

4. Addition of alcohol-

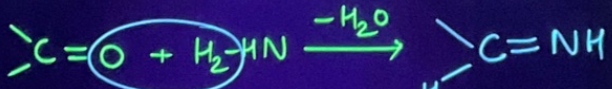
Aldehydes react with one equivalent of monohydric alcohol in the presence of dry hydrogen chloride to yield alkoxyalcohol intermediate, known as hemiacetals, which further react with one more molecule of alcohol to give a gem-dialkoxy compound known as acetal.



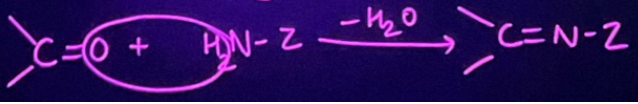


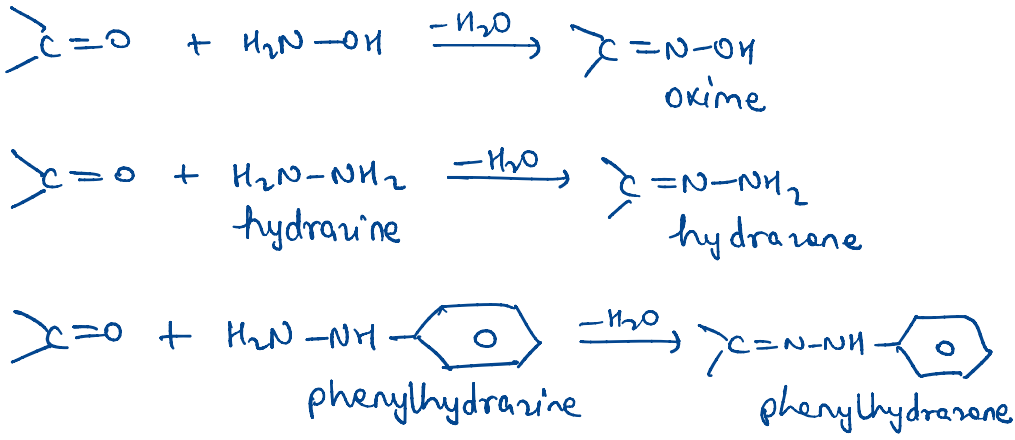
5. Addition of ammonia & its derivatives- ✓

Nucleophiles, such as ammonia and its derivatives H2N-Z add to the carbonyl group of aldehydes and ketones. The reaction is reversible and catalysed by acid. The equilibrium favours the product formation due to rapid dehydration of the intermediate to form >C=N-Z.



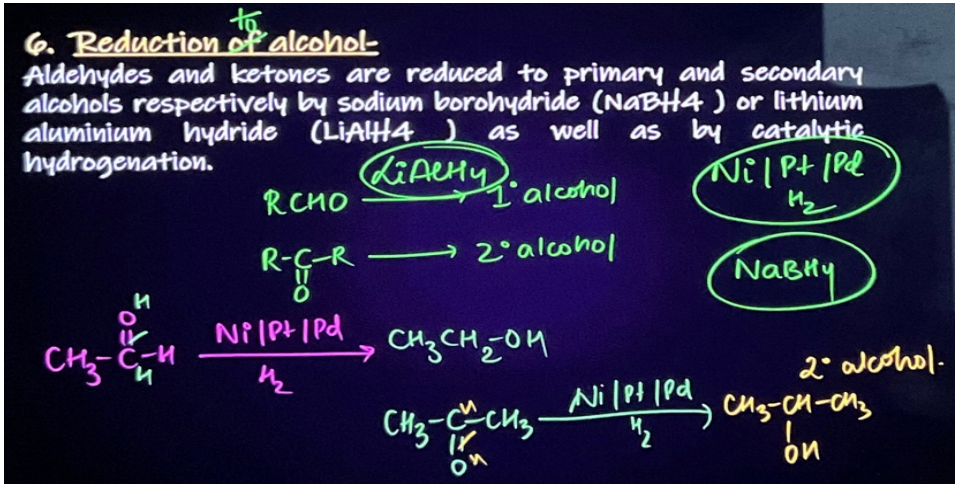
NH_2-(Z)





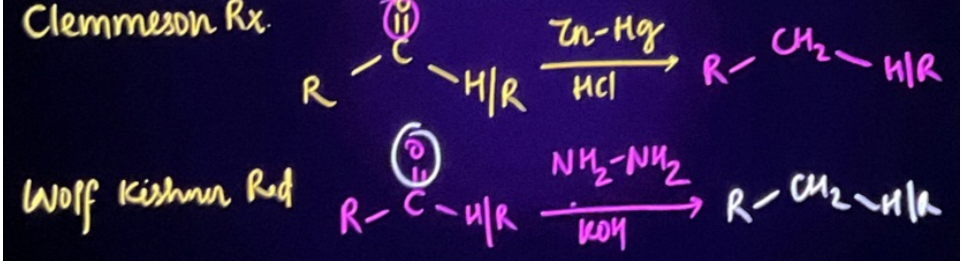
6. Reduction of alcohol-

Aldehydes and ketones are reduced to primary and secondary alcohols respectively by sodium borohydride ($NaBH_4$) or lithium aluminium hydride ($LiAlH_4$) as well as by catalytic hydrogenation.



7. Reduction of hydrocarbons-

The carbonyl group of aldehydes and ketones is reduced to CH_2 group on treatment with zincamalgam and concentrated hydrochloric acid reduction] or with hydrazine followed by heating with sodium or potassium hydroxide in high boiling solvent such as ethylene glycol (Wolff-Kishner reduction).



8. Oxidation-

$$RCHO \xrightarrow[K_2Cr_2O_7]{KMnO_4} RCOOH$$

9. Aldol condensation reaction-

$-H_2O$ → enal.

ald + ol

aldehyde alcohol.

Self aldol condensation

2 Molecule & same

→ dil NaOH ($\bar{O}H$ act as base)

→ α-hydrogen should be present.

Cross aldol Condensation

2 Molecule & diff.

accept H^+

Self Aldol condensation.

$$CH_3-\overset{O}{\overset{\|}{C}}-H + CH_3-\overset{O}{\overset{\|}{C}}-H \xrightarrow{dil.\ NaOH}$$

occurs in 2 steps -

1) formation of carbanion (1st molecule)

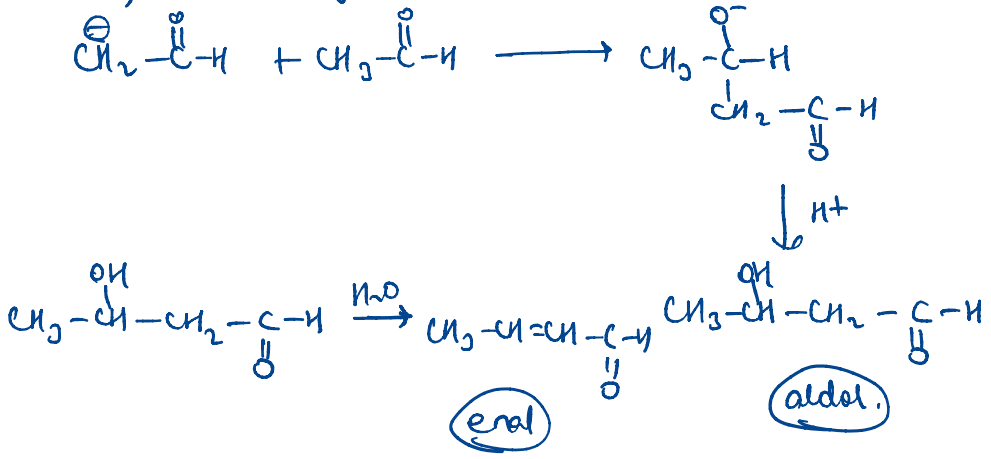
2) Attack of carbanion on 2nd molecule.

1) Formation of Carbanion.

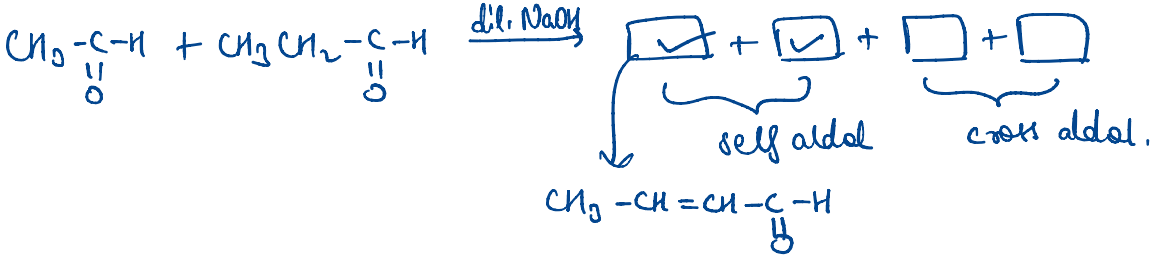
$$CH_3-\underset{\alpha\text{-}C}{C}(=O)-\underset{\alpha\text{-}H}{H} \xrightarrow[\bar{O}H]{dil.\ NaOH} CH_2-C(=O)-H \longrightarrow \overset{\ominus}{C}H_2-C(=O)-H$$

carbanion.

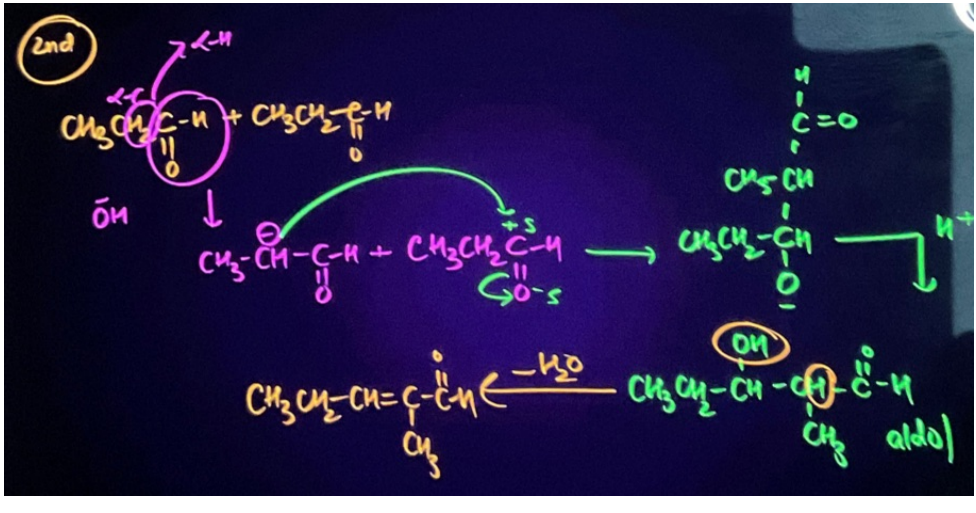
2.) attack of carbanion on second molecule.



cross aldol condensation



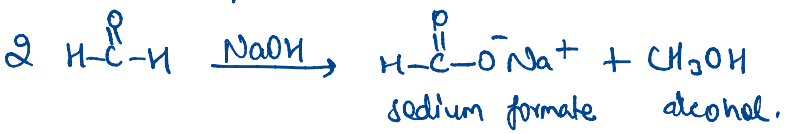
2nd – 1st $C^{\ominus}$ → 2nd attack
4th → 2nd $C^{\ominus}$ → 1st attack



Cannizaro Rx –

↓

no. α-H required



Carboxylic Acid

↓

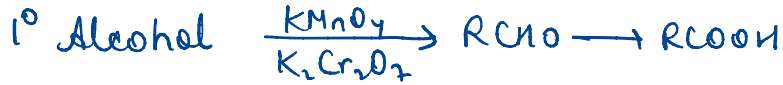
RCOOH or 

2° suffix = oic acid or carboxylic acid

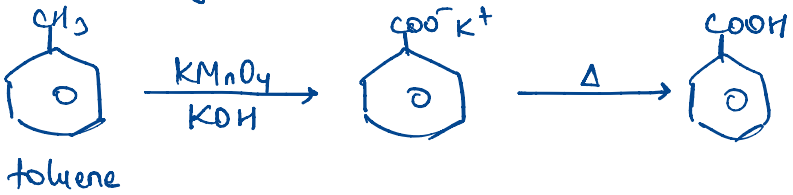
2° prefix = carboxy.

MOP of Carboxylic Acid

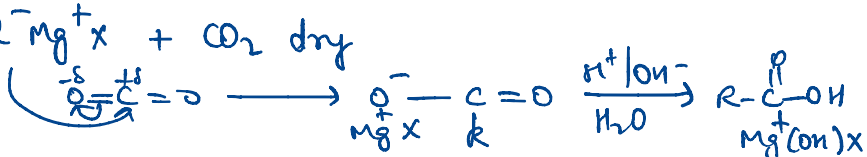
1.) Oxidⁿ of alcohol



2.) From alkyl benzene



3.) From grignard Reagent



4.) From nitriles

$$R-C\equiv N \xrightarrow{H_2O} R-\underset{\underset{O}{\|}}{C}-NH_2 \xrightarrow{H_2O} R-\underset{\underset{O}{\|}}{C}-OH + NH_3$$

5.) Ozonolysis

$$R-CH=CH_2 \xrightarrow[Zn]{O_3} RCOOH + HCOOH$$

(oxidative ozonolysis)

6.) $R-\underset{\underset{O}{\|}}{C}-Cl \xrightarrow{H_2O} R-\underset{\underset{O}{\|}}{C}-OH$

Chemical properties

a) acidity

$$R-\underset{\underset{O}{\|}}{C}-OH \xrightarrow{-H^+} R-\underset{\underset{O}{\|}}{C}-O^- \quad (\text{anion})$$

$\xrightarrow[EDG]{EWG}$ $\propto$ acidity $\propto$ anion order $\propto \frac{-I\ \alpha\ -M}{+I\ \propto\ +M}$

which is more stable

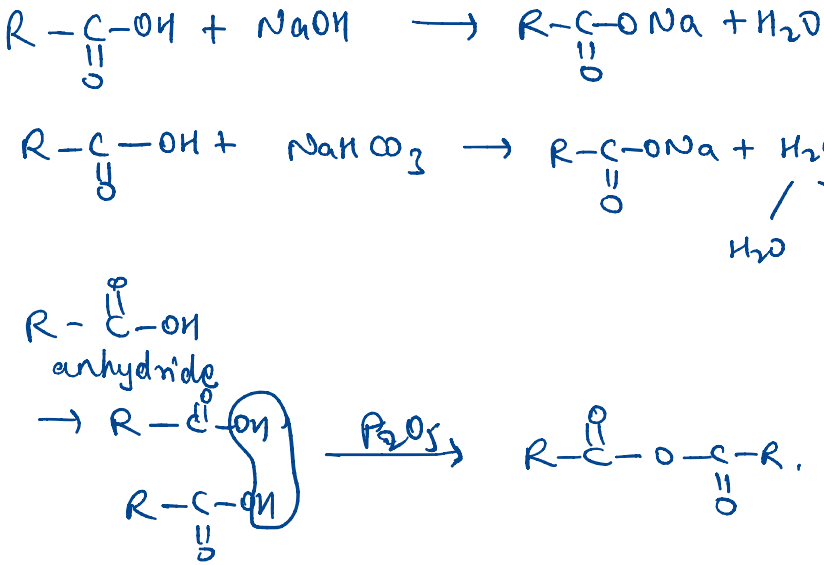
Carboxylic acid > Phenol

$R-\underset{\underset{O}{\|}}{C}-OH$ > 

resonance
eg. resonance

Resonance

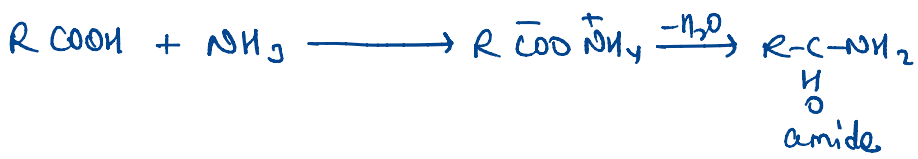
$$R-\underset{\underset{O}{\|}}{C}-OH + Na \longrightarrow R-\underset{\underset{O}{\|}}{C}-ONa + H_2$$



Esterification



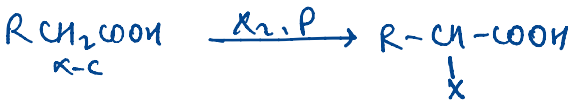
Rx with NH_3



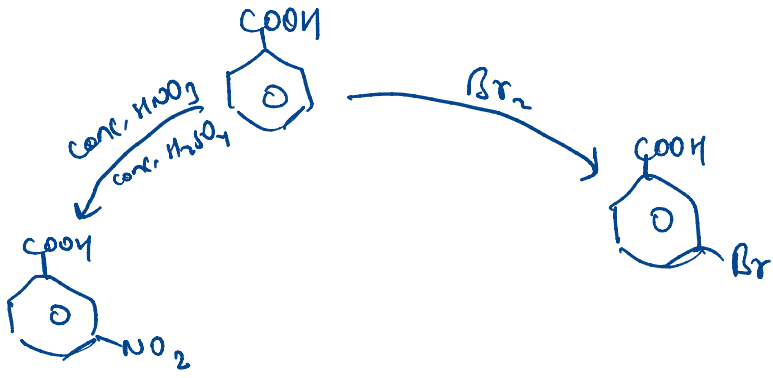
Decarboxylation



HVZ Rx

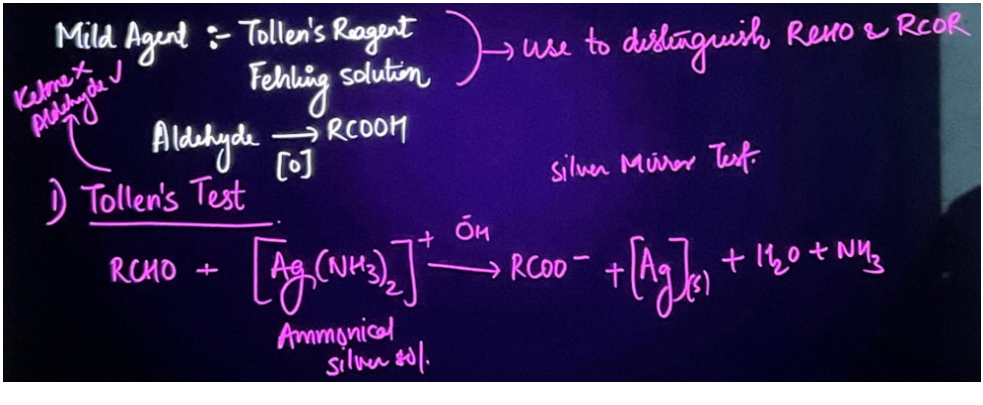


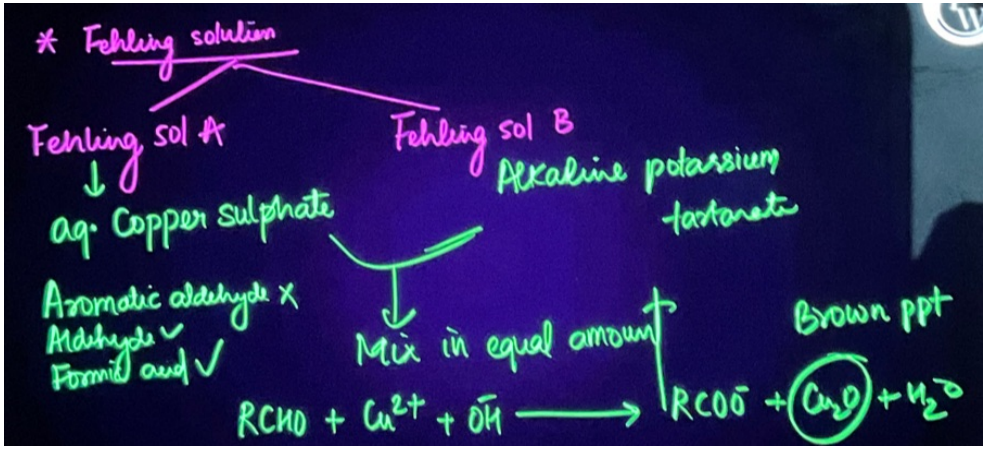
Ring Substitution –

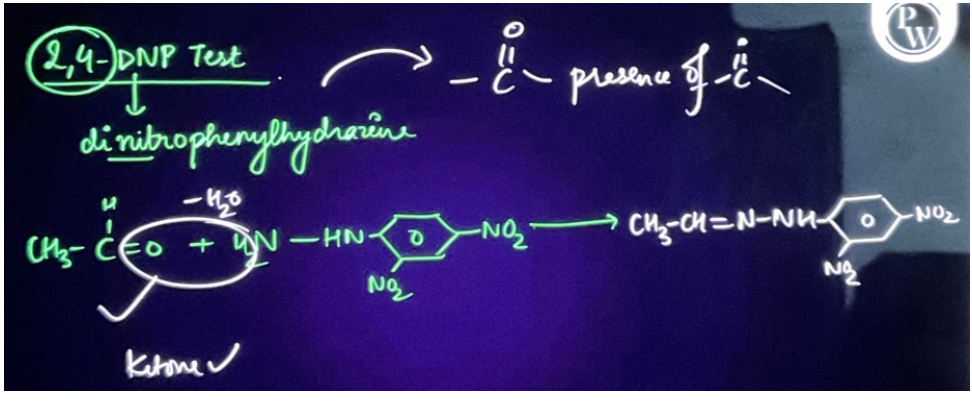


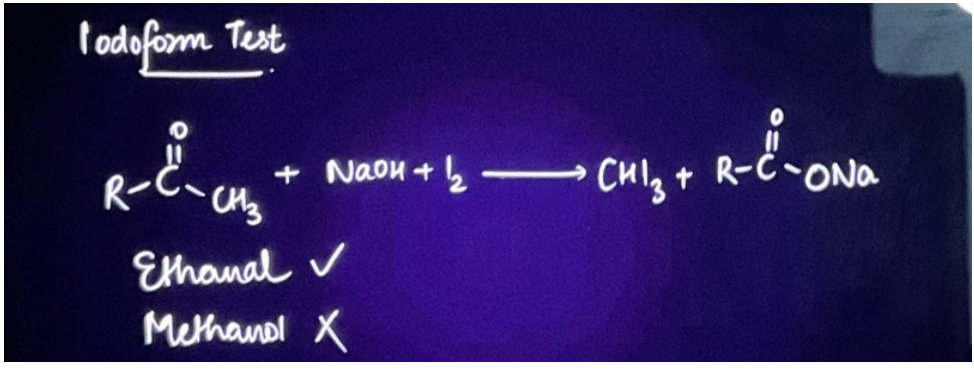
2,4-Dinitrophenyl Hydrazine Test	The formation of orange-yellow crystals indicates the presence of the carbonyl group
Sodium Bisulphite Test	The formation of crystalline precipitate confirms the carbonyl group.
Schiff's Test	The appearance of pink, red or magenta colour indicates the presence of the aldehyde group.
Fehling's Test	The appearance of red precipitate confirms the presence of an aldehydic group.
Tollen's Test (Silver Mirror Test)	The appearance of a shiny silver mirror confirms the presence of aldehydes.
Test with Chromic Acid	The appearance of green or blue colour precipitate confirms the presence of aldehydes.
Sodium Nitroprusside Test	The appearance of red colouration shows the presence of ketone.

Distinction Test




* Fehling solution
Fehling sol A
Fehling sol B
aq. Copper Sulphate
Alkaline potassium tartarate
Aromatic aldehyde ✗
Aldehyde ✓
Formic acid ✓
Mix in equal amount
Brown ppt
RCHO + Cu2+ + OH⁻ ⟶ RCOO⁻ + Cu2O + H2O


2,4-DNP Test
dinitrophenylhydrazine
presence of -C(=O)-
Ketone ✓


Iodoform Test
NaOH + I2
CHI3
Ethanal ✓
Methanol ✗

THANK YOU!!